

# KISHOR KUMAR PAROI


**AI Systems Architect | (Dev/ML/LLM/AI)Ops | Founder@KoomAI | FDE(LangGraph, Kubernetes, Kubeflow, GCP, Hybrid RAG<35ms, Agent Mesh, AI Security) | vLLM, Finetune | Data Engineer(Azure, AWS, Databricks, Snowflake) | Rust SIMD**

**Founder @ KoomAI (koom | GitHub)** • B.Sc. in CSE @ RUET • Dhaka, Bangladesh / Global Remote

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| Year | Degree / Certificate | Institute / Board | CPI / Marks | Merit Rank |
|---|---|---|---|---|
| 2018 -- 2023 | B.Sc. in Computer Science & Engineering | Rajshahi Univ. of Eng. & Tech. (RUET) | **3.25 / 4.00 (D.O)** | Rank 483rd in Bangladesh |
| 2015 -- 2017 | Higher Secondary Certificate (HSC) | Saint Joseph Higher Secondary School (Dhaka) | **5.00 / 5.00** | Rank 85th in Dhaka Board |
| 2013 -- 2015 | Secondary School Certificate (SSC) | S.J Govt Model High School (Gopalganj) | **5.00 / 5.00** | Rank 90th in Dhaka Board |

## WORK EXPERIENCE

**Omnichannel AI — Voice & Chat Receptionist** *Sept'26 – Present*
*Lead Voice AI Architect • PSTN: +1-617-271-8180 • 6-Vertical Multi-Tenant SaaS*
Live | WebRTC | GitHub

- Launched live **6-vertical Vapi Squad (Marcus↔Sophia)** on **Deepgram Nova-2 STT (<180ms), FastAPI (<40ms), Cartesia Sonic-3.5 (<110ms)**; sub-500ms PSTN, WebRTC, WhatsApp, IG, FB & Telegram.
- Built **Clerk + Neon PostgreSQL + Google Calendar v3** backend with **n8n/Slack** escalation, vector FAQ RAG & **Twilio SMS** 24h/1h cron reminders.

**phi0.ai** *Jul'26 -- Oct'26*
*Forward Deployed AI Engineer (Production RAG, GenAI & MLOps, Europe | Remote)*

- Orchestrated forward deployment of **5 concurrent autonomous AI agents** executing multi-step tools via **LangGraph**; attained **99.4% task completion**, accelerating client onboarding by **4x**.
- Devised high-throughput **Hybrid RAG** with HNSW, BM25, and multi-stage re-ranking (**FlashRank + Cohere v3.5**) under **<35ms semantic cache**; curtailed hallucinations by **46%** and saved **$18K/mo**.
- Codified continuous **MLOps & GenAI flywheel** with NeMo Guardrails and Canary K8s deployments; spearheaded enterprise AI transformation, securing **$1.4M in client contracts**.

**Samsung AI Core / Care Health** *Mar'26 -- Jun'26*
*Data Engineering & AI Systems Intern*

- Formulated batch & streaming pipelines via **AWS S3, EMR, Athena, PySpark**; slashed operational compute overhead by **$4.2K/run (34% savings)** through asynchronous event-driven Lambda/SQS dispatch.
- Accelerated 8B/70B model inference on H100 GPUs via **FlashAttention-2** and fused kernels; delivered **3.8x throughput** on >32K context windows, mitigating p99 latency by **54%**.
- Deployed real-time health insurance claim fraud detection pipeline over **AWS Kinesis and OpenTelemetry**; eradicated false-positive claim investigations by **52%**, preserving **$380K annually**.

**Independent AI Specialization & Sabbatical** *Oct'25 -- Feb'26*
*Full-Time Technical Deep Dive: Math, ML/DL, GenAI, Big Data & MLOps*

- Rigorous hands-on mastery in **Linear Algebra** (SVD, PCA), **Calculus**, and core ML; handcrafted Transformers, Self-Attention, and CNNs from first principles in **PyTorch**.
- Devised **Hybrid RAG**, **LangGraph agents**, **Big Data Lakehouses** (PySpark Delta Lake, Snowflake, Kafka streams, dbt Core), and MLOps flywheels (**MLflow, Feast, K8s**).

**kepi.ai** *Jan'22 -- Sep'25*
*Senior Full Stack Engineer (3 yrs 9 mos, Remote)*

- Architected zero-downtime microservices migration to **Kubernetes on AWS**; scaled platform throughput by **300%** while curtailing p95 latency by **48%** at 99.99% availability.
- Spearheaded enterprise AI initiatives: delivered autonomous multi-agent pipelines and scalable LLM microservices, driving a **22% increase in customer LTV**.
- Fortified zero-trust mesh infrastructure with **Istio mTLS and fine-grained RBAC**; precluded unencrypted service-to-service transit and diminished cross-zone egress costs by **38%**.

## FOUNDER VENTURE: KOOMAI

**KoomAI — Async Video Intelligence SaaS** *Jan'25 -- Dec'25*
*Founder & Principal Architect | Desktop, MV3 Extension, Express & Next.js on AWS*
Live | Next.js | Desktop | Extension | Backend

- **WebRTC 60 FPS & Desktop:** Sub-120ms WebRTC pipeline + Electron NVENC recorder; 60 FPS 1080p across **50K+ video mins** (-45% CPU).
- **20-Agent Mesh & Copilot:** Deployed 20 **LangGraph** agents (**<35ms p99**) generating 1-click GitHub PRs, Jira stories & Playwright tests.
- **NotebookLLM & AI Recon:** Two-host audio podcast studio, system design drills & automated **ai-job-search** tailored ATS CV engine.
- **Chrome MV3 & Invoicing:** Screen recorder with Pomodoro sprints, 432Hz audio & instant **Stripe invoicing** ($24K MRR).

## ACHIEVEMENTS & HACKATHONS

- **Competitive Programming:** **232nd** Codeforces Div-2 (**Blue Coder**), 5★ CodeChef, **17th IUPC**; **Meta Hacker Cup R3**; 2000+ DSA problems.
- **Champion | Technobeast AI Agents Hackathon:** Qdrant negative-embedding optimization & transformer trajectory matching.
- **7th Rank | AI Security Hackathon:** Real-time prompt injection firewall (**97.2% accuracy**, **<30ms** latency).
- **National Merit Scholarships:** 85th Rank Dhaka Board (HSC) and 90th Rank Dhaka Board (SSC).

## CERTIFICATIONS & KEY COURSES

- **DeepLearning.AI / Stanford (Andrew Ng):** ML & Deep Learning Specializations (NNs, CNNs, Transformers, Embeddings).
- **Systems & Data Architecture:** Distributed Systems & AI Design (Arpit Bhayani, Gaurav Sen, ByteByteGo), Lakehouses (Zach Wilson, Mayank Agarwal), MLOps (Krish Naik).
- **Core CS (RUET):** DSA, OOP (C++/Java), OS, DBMS, Distributed & Cloud Computing, Networks, Compilers, Automata, Software Eng., Architecture, AI, Microprocessors, Discrete Math, Network Security & Cryptography.

## LEADERSHIP & MENTORING

- **AI Agents Masterclass & Mentorship:** Led in-person **LangGraph** multi-agent workshops and mentored **100+ engineers** across ML, AI Agents, DevOps & MLOps through code reviews and system design.

## TECHNICAL PROJECTS

One-Click Deploy: curl | bash

**Zomato AI Lakehouse & Multimodal Platform** Deploy *Sept'26 -- Present*
*Snowflake Delta Lakehouse (35M+ Rows), Feast Store & CV Engine*

- Architected **Snowflake & PySpark Delta** Lakehouse on **35M+ orders (2.3 GB)** with **dbt Core SCD-2**, cutting compute **42% ($18K/mo)**.
- Streamed real-time features via **Airflow, Debezium CDC WAL** & Redis **Feast Store** (**<5ms** serving), unlocking **+$340K/qtr** lift.
- Built multimodal CV fraud arbitration (**dHash**, OCR, **G-Eval CoT**), Delta compaction & DR (**RPO<60s, RTO<4.5m**); cut fraud **68% ($1.2M/yr saved)**.

**Hybrid Anime Intelligence & LLMOps Platform** Deploy *Aug'26 -- Present*
*Polyglot Mesh (Go/Rust), RSLoRA Flywheel & Multi-LoRA vLLM*

- Polyglot mesh (**Go Gateway, Rust SIMD, FastAPI, Kafka**) cut vector scoring from 25ms to **<0.5ms (50x)**, saving **64% CPU**.
- Combined **Dual-Tower NCF, ChromaDB HNSW & BM25** via RRF (**NDCG@10 = 0.94**) + **FlashRank ONNX (+21.4% MRR)**; +28% retention.
- Served **vLLM Multi-LoRA** + 4-bit AWQ & **RSLoRA flywheel** with KEDA (420 users/GPU); cut GPU spend **68%** at **3.4x throughput**.

**Nexus-AI Multi-Cloud Lakehouse & Agent Mesh** Deploy *Oct'26 -- Present*
*Multi-Cloud (AWS/GCP/Azure), LangGraph, Delta Lake & Next.js 16*

- Orchestrated **LangGraph** multi-agent mesh (NL2SQL, BI, MLOps, OWASP) with **MLflow, Feast & K8s**; **99.2% reliability**, <12m deploy.
- Unified **AWS S3, GCP GCS & Azure ADLS Gen2** via **PySpark Delta & dbt** (18M+ events, **<250ms**); Go (:3000)/FastAPI/Next.js 16 SSE (-65% overhead).
- Enforced zero-trust defense via **NeMo Guardrails, BotShield WAF & Terraform IaC**; blocked **98%+ injections** with 1-click bootstrap.

**Enterprise AI Security & Agentic RAG Platform** Deploy *Oct'26 -- Present*
*Zero-Trust AI Gateway, NVIDIA NeMo & LangGraph*

- Hardened gateway with **BotShield WAF, NeMo Guardrails & PyRIT 0.14.0**; blocked **95%+ jailbreaks** & cut prompt tokens **36%**.
- Built stateful **LangGraph + Portkey Gateway** core with **<50ms semantic cache**; saved **$8.4K/mo (42%)** with RAGAS eval harness.
- Standardized **FastMCP + HITL gates**, HMAC-SHA256 tool signing & Logfire/LangSmith tracing; achieved **100% audit compliance**.

**Smart Manufacturing Industrial AI & MLOps** Deploy *Oct'26 -- Present*
*Rust/Go/Next.js, PyTorch Autoencoders, GKE/K3s & ArgoCD*

- Deployed **PyTorch Autoencoder + CNN (<20ms)** & **TensorRT INT8 (<8.2ms)** inspection; cut escape defects **31% ($420K/yr scrap saved)**.
- Built polyglot edge: **Rust SIMD** (:8082), **Go Gateway** (:8085), **OPC-UA (50K tags/s)**, gRPC (-68% BW) & Next.js 15 (:3000).
- Automated CI/CD via **Jenkins, SonarQube, Nexus 3 & ArgoCD GitOps**; slashed deploy cycle **96% (4h → 8m)** with zero-downtime failover.

**Core Systems: Rust Git Engine & POSIX Shell** *Aug'24*
*Bare-Metal Rust Systems, Concurrency & Pipelines*

- Crafted Git internals in **Rust** (SHA-1 object graph, zlib SIMD); matched C throughput with **100% compile-time memory safety**.
- Built async POSIX shell with signal multiplexing & I/O redirection; achieved **<1µs latency** and **73% lower heap allocation**.

**FlashAttention-2, 3 & Flash-Decoding Kernel** *Oct'26 -- Present*
*Bare-Metal CUDA/C++, Split-KV, PagedAttention & FP8*

- Handcrafted **CUDA FlashAttention-2/3** eliminating $O(N^2)$ HBM via Tiled Online Softmax; **Split-KV Flash-Decoding** hit **100% SM occupancy (8x lower decode latency)**.
- Engineered **PagedAttention** (<4% waste), fused ALiBi sliding-window & FP8 kernels, plus O(N) SRAM-recompute backward pass (39 PyTorch equivalence suites).

**Distributed LLM Training (4D Parallelism)** *Oct'26 -- Present*
*4D Parallelism (TP×PP×DP×SP), Interleaved 1F1B & ZeRO-3*

- Handcrafted LLaMA-3 (8B/70B) framework with **Megatron TP, Interleaved 1F1B PP, ZeRO-3 FSDP & Ring Attention** (1M+ tokens, 100% overlap) + DCP resharding.
- Implemented **ZeRO-3 parameter lifecycle** & custom **Ring All-Reduce** ($2((P-1)/P)NB$); sustained **49.07% MFU across 64x H100s**.

**AutoCut-AI: Multi-Modal Video Editor** *Aug'26 -- Present*
*OpenAI Whisper Diarization, FFmpeg GPU NVENC & Next.js*

- Built Whisper diarization, silence/filler removal & FFmpeg NVENC animated subtitles; creates 9:16 reels in **<45s (4.2x retention lift)**.

**Nimbus-AI: 5-Agent Live Voice Suite** Live *Sept'26 -- Present*
*Live Agents: Sales () · Meeting () · Research () · Schedule () · Desktop ()*

- Built scratch **5-agent voice platform** on **LiveKit WebRTC, FastAPI, GPT-4o/Whisper & ElevenLabs Turbo v2.5** achieving **<400ms** roundtrips.
- Engineered **60fps audio waveform**, instant barge-in interruption, RAG vs RAG-less routing & 5 production specialist tools (CRM, Scribe, Citations, Calendar, OS).

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, C/C++, Rust (Actix), Go (Gin), SQL, Bash, TypeScript, Java (Spring Boot), FastAPI, Express, NestJS, Next.js 16, Streamlit
**AI, Voice & LLMOps:** PyTorch, vLLM, Vapi, Deepgram Nova-2, Cartesia Sonic-3.5, LangGraph, LangChain, HF, TRL, RSLoRA/QLoRA, MLflow, Feast, Kubeflow, SageMaker, KServe, DVC
**Data & Lakehouse:** Snowflake, Databricks, Azure Data Factory (ADF), Synapse CETAS, ADLS Gen2, dbt Core, PySpark, Delta Lake, Kafka, Airflow, Debezium CDC
**Cloud & Infra:** AWS, GCP, Azure, Terraform, Ansible, Helm, ArgoCD, Sonatype Nexus, Docker, Kubernetes (GKE, EKS, K3s)
**DevSecOps & Obs:** NVIDIA NeMo Guardrails, PyRIT, BotShield WAF, SonarQube, Trivy, Jenkins, GitLab CI/CD, Prometheus, Grafana, EFK (Elasticsearch/Kibana)
**WebRTC, 3D & Full-Stack:** WebVR, WebRTC, Three.js, R3F, ElectronJS, Next.js 16, Clerk, n8n, Twilio, Neon PostgreSQL, Prisma, Drizzle, Stripe Connect, MongoDB Atlas